

PrimePy: Number Theory Library- User Guide

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1 Introduction

This user guide provides a practical overview of how to use the **PrimePy** library. Rather than focusing on the underlying mathematical theory or algorithmic design, this document is dedicated to demonstrating how to work with the library from a programmer's perspective.

Through concise and focused examples, users will learn how to import and interact with the various components of the library, including core number theory utilities, algebraic structures, and cryptographic primitives. Each section includes annotated code snippets and explanations to help both beginners and advanced users integrate **PrimePy** into their own projects.

For the theoretical background and detailed explanation of the algorithms implemented, please refer to the accompanying document.

2 Algebra

This section introduces the core abstract algebraic structures supported by the **PrimePy** library. We begin with a brief overview of each structure and then provide concrete examples of their implementations within the library.

The main structures currently supported are:

1. **Groups**
2. **Rings**